SESD-Assignment-1

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Monday – Case Study 1: Online Library Management System

Description:
A web-based application for students to search, borrow, and return books from a university library.

Sample Functional Requirements:

- Students can register and log in
- Search for books by title/author
- Place a hold or borrow books
- Return or renew borrowed books
- Admin can add/remove/update books

Sample User Stories:

- As a student, I want to search for books so I can find what I need.
- As a student, I want to borrow a book online so I can read it at home.
- As an admin, I want to add new books so the catalog is up-to-date.

Suggested Epics:

- User Account Management
- Book Search and Borrowing
- Admin Book Inventory

1. Description

The University Library Management System is a web-based application that allows students to search, borrow, return, and renew library books online. The admin manages book records and monitors library activities.

2. Functional Requirements

1. Students can register and log in to the system.
2. Students can search for books by title or author.
3. Students can check book availability.
4. Students can borrow available books.
5. Students can place a hold on unavailable books.
6. Students can return borrowed books.
7. Students can renew books before the due date.
8. Students can view their borrowing history.
9. Admin can add, update, and remove books.
10. Admin can monitor borrowed and returned books.
11. The system sends notifications about due dates.

12. The system calculates fines for late returns.

3. User Stories

1. As a student, I want to register an account so that I can access library services.
2. As a student, I want to log in so that I can manage my borrowed books.
3. As a student, I want to search for books so that I can find the books I need.
4. As a student, I want to check book availability so that I know whether a book can be borrowed.
5. As a student, I want to borrow books online so that I can read them at home.
6. As a student, I want to reserve unavailable books so that I can borrow them later.
7. As a student, I want to renew borrowed books so that I can extend the borrowing period.
8. As a student, I want to receive due-date reminders so that I can return books on time.
9. As an admin, I want to add new books so that the library catalog stays updated.
10. As an admin, I want to update or remove book records so that the inventory remains accurate.
11. As an admin, I want to track borrowed books so that I can monitor library activities.
12. As an admin, I want to manage overdue books so that library rules are followed.

4. Epics

Epic 1: User Account Management

- Student registration
- Student login and logout
- Profile management
- Password reset
- Account authentication

Epic 2: Book Search and Borrowing

- Search books by title or author
- Check book availability
- Borrow books
- Reserve unavailable books
- Return borrowed books
- Renew borrowed books
- View borrowing history
- Receive due-date notifications

Epic 3: Admin Book Inventory

- Add new books

- Update book information
- Remove books
- Manage book availability
- Track borrowed and returned books
- Monitor overdue books
- Generate library reports

5. Non-Functional Requirements

1. Security: User information must be protected.
2. Performance: Book searches should return results quickly.
3. Usability: The system should be simple and user-friendly.
4. Availability: The application should be accessible whenever required.
5. Reliability: Book records and transactions should be accurate.
6. Scalability: The system should support increasing numbers of students and books.

6. Conclusion

The University Library Management System simplifies library operations by allowing students to search, borrow, return, and renew books online. It also helps administrators maintain book inventory and monitor library activities efficiently.

